
#Task 1

```
import os

filename = "diary.txt"

print("Checking if diary file exists...")

if not os.path.exists(filename):
    print("Diary not found, creating new diary\n")
    with open(filename, "w") as f:
        f.write("Day 1: Started learning Python\n")
        f.write("Day 2: Practiced file handling\n")
        f.write("Day 3: Learned conditions\n")
        f.write("Day 4: Worked on loops\n")
        f.write("Day 5: Feeling confident\n")

print("Diary content:")
with open(filename, "r") as f:
    content = f.read()
    print(content)

print("First 50 characters:")
print(content[:50])

print("\nReading diary line by line:")
with open(filename, "r") as f:
    for line in f:
        print(line.strip())

os.remove(filename)
print("\nDiary deleted")
```

RESTART C:/Users/ADMIN/Desktop

```
Checking if diary file exists...
Diary not found, creating new diary
```

```
Diary content:
Day 1: Started learning Python
Day 2: Practiced file handling
Day 3: Learned conditions
Day 4: Worked on loops
Day 5: Feeling confident
```

```
First 50 characters:
Day 1: Started learning Python
Day 2: Practiced fi
```

```
Reading diary line by line:
Day 1: Started learning Python
Day 2: Practiced file handling
Day 3: Learned conditions
Day 4: Worked on loops
Day 5: Feeling confident
```

```
Diary deleted
```

```
#Task 2
```

```
print("Mobile Recharge Recommendation System")

mobile = input("Enter mobile number: ")
budget = int(input("Enter budget amount: "))

print("Checking suitable plan...")

if budget == 199:
    print("Recommended Plan: ₹199 for 28 days")
elif budget == 299:
    print("Recommended Plan: ₹299 for 56 days")
elif budget == 499:
    print("Recommended Plan: ₹499 for 84 days")
else:
    print("No suitable plan available")
```

```
===== RESTART: C:/Users/krsu/OneDrive
Mobile Recharge Recommendation System
Enter mobile number: 6363339707
Enter budget amount: 499
Checking suitable plan...
Recommended Plan: ₹499 for 84 days
|
```

#Task 3

```
print("Washing Machine Load Manager")

cloth_type = input("Enter cloth type: ")
count = int(input("Enter number of clothes: "))
weight = float(input("Enter weight per cloth: "))

total_weight = count * weight

print("Calculating total load weight...")

if total_weight <= 7:
    status = "Wash Started"
else:
    status = "Overload! Reduce clothes"

print("Cloth Type:", cloth_type)
print("Total Load Weight:", total_weight, "kg")
print("Washing Status:", status)
|
```

```
----- RESTART: C:/Users/krsu/OneDrive
Washing Machine Load Manager
Enter cloth type: cotton
Enter number of clothes: 12
Enter weight per cloth: 0.5
Calculating total load weight...
Cloth Type: cotton
Total Load Weight: 6.0 kg
Washing Status: Wash Started
```

#Task 4

```
class BusTicket:
    def __init__(self, name, bus_no, seat_no):
        self.name = name
        self.bus_no = bus_no
        self.seat_no = seat_no

print("Creating bus ticket objects")

t1 = BusTicket("Asha", "KA05MX0846", 12)
t2 = BusTicket("Ravi", "KA20GH9089", 18)

print("Passenger 1:", t1.name, t1.bus_no, t1.seat_no)
print("Passenger 2:", t2.name, t2.bus_no, t2.seat_no)

===== RESTART: C:/Users/krsb/OneDrive
Creating bus ticket objects
|Passenger 1: Asha KA05MX0846 12
Passenger 2: Ravi KA20GH9089 18
```

#Task 5

```
from abc import ABC, abstractmethod

class Mobile(ABC):
    def __init__(self, pin):
        self.__pin = pin

    def verify_pin(self, p):
        return p == self.__pin

    def change_pin(self, old, new):
        if self.verify_pin(old):
            self.__pin = new
            print("PIN changed successfully")
        else:
            print("Incorrect PIN")

    @abstractmethod
    def lock(self):
        pass

class Phone(Mobile):
    def lock(self):
        print("Phone is locked")

print("Initializing mobile security system")

m = Phone(1234)
print("Verifying PIN:", m.verify_pin(1234))
m.change_pin(1234, 5678)

----- RESTART: C:/Users/KISAW/OneDr...
Initializing mobile security system
Verifying PIN: True
PIN changed successfully
```

#Task 6

```
import datetime
import math
import os

print("Smart Travel Planner")

name = input("Enter customer name: ")
city = input("Enter destination city: ")
travelers = int(input("Enter number of travelers: "))
cost = float(input("Enter cost per ticket: "))

total = travelers * cost
final_cost = math.ceil(total + total * 0.05)

print("Calculating total cost with service charge...")

time = datetime.datetime.now()
file = "booking.txt"

if not os.path.exists(file):
    with open(file, "w") as f:
        f.write(name + "\n")
        f.write(city + "\n")
        f.write(str(final_cost))

print("Booking Time:", time)
print("Final Cost:", final_cost)
print("Booking file created")

===== RESTART: C:/Users/KrSub/OneDrive/Desktop
Smart Travel Planner
Enter customer name: Rajesh
Enter destination city: Mysore
Enter number of travelers: 4
Enter cost per ticket: 399
Calculating total cost with service charge...
Booking Time: 2026-02-22 20:38:08.209411
Final Cost: 1676
Booking file created
.
```

```

#Task 7

import datetime
import os
import sys

print("Smart Health Check System")

name = input("Enter patient name: ")
age = int(input("Enter age: "))
temp = float(input("Enter body temperature: "))
hr = int(input("Enter heart rate: "))

status = "Fever Detected" if temp > 37.5 else "Normal"

time = datetime.datetime.now()
file = "health_report.txt"

if not os.path.exists(file):
    with open(file, "w") as f:
        f.write(name + "\n")
        f.write(status + "\n")

print("Health Status:", status)
print("Checkup Time:", time)
print("Python Version:", sys.version)

```

```

===== RESTART: C:/Users/krsb/OneDrive/Desktop/internship/DAY 18 TASK.py =====
Smart Health Check System
Enter patient name: Kumar
Enter age: 24
Enter body temperature: 47
Enter heart rate: 98
Health Status: Fever Detected
Checkup Time: 2026-02-22 20:40:04.970116
Python Version: 3.10.0 (tags/v3.10.0:b494f59, Oct 4 2021, 19:00:18) [MSC v.1929
64 bit (AMD64)]

```

#Task 8

```
import math
import datetime
import os

print("Smart Weather Analyzer")

temps = []
for i in range(5):
    temps.append(float(input("Enter temperature: ")))

avg = sum(temps) / 5

print("Calculating weather condition...")

if avg >= 35:
    result = "Hot Week"
elif avg >= 20:
    result = "Pleasant Week"
else:
    result = "Cold Week"

file = "weather_report.txt"
if not os.path.exists(file):
    with open(file, "w") as f:
        f.write(result)

print("Average Temperature:", math.floor(avg))
print("Weather Result:", result)
print("Report Date:", datetime.date.today())
```

```
===== RESTART: C:/Users/krsu/OneDrive
Smart Weather Analyzer
Enter temperature: 33
Enter temperature: 23
Enter temperature: 27
Enter temperature: 18
Enter temperature: 35
Calculating weather condition...
Average Temperature: 27
Weather Result: Pleasant Week
Report Date: 2026-02-22
|
```

```

#Task 9

import tkinter as tk

def add():
    result.config(text="Result: " + str(int(e1.get()) + int(e2.get())))

def sub():
    result.config(text="Result: " + str(int(e1.get()) - int(e2.get())))

def mul():
    result.config(text="Result: " + str(int(e1.get()) * int(e2.get())))

def div():
    result.config(text="Result: " + str(int(e1.get()) / int(e2.get())))

root = tk.Tk()
root.title("Simple Calculator")

e1 = tk.Entry(root)
e2 = tk.Entry(root)
e1.pack()
e2.pack()

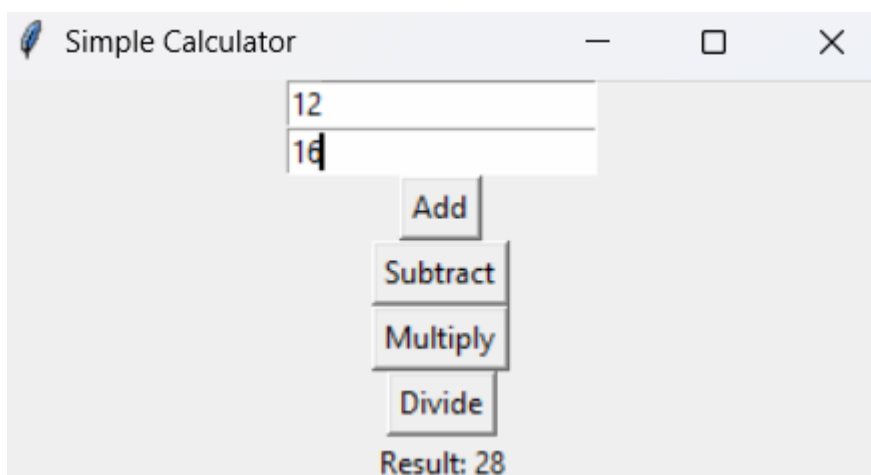
tk.Button(root, text="Add", command=add).pack()
tk.Button(root, text="Subtract", command=sub).pack()
tk.Button(root, text="Multiply", command=mul).pack()
tk.Button(root, text="Divide", command=div).pack()

result = tk.Label(root, text="Result")
result.pack()

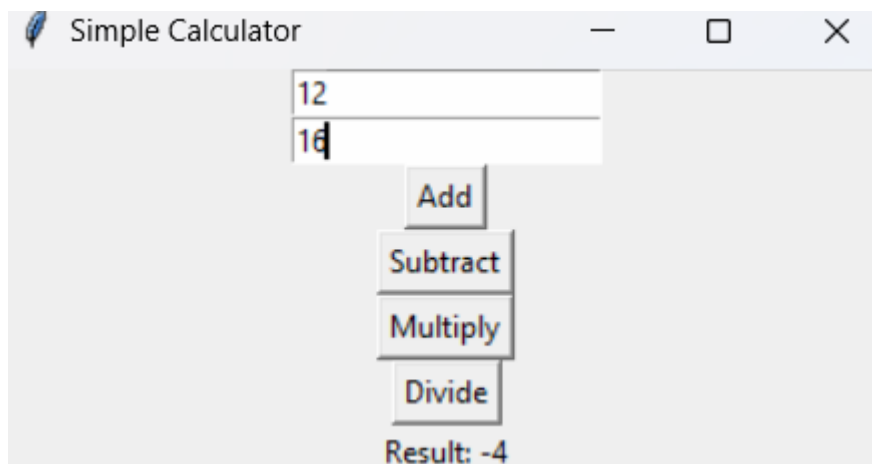
root.mainloop()

```

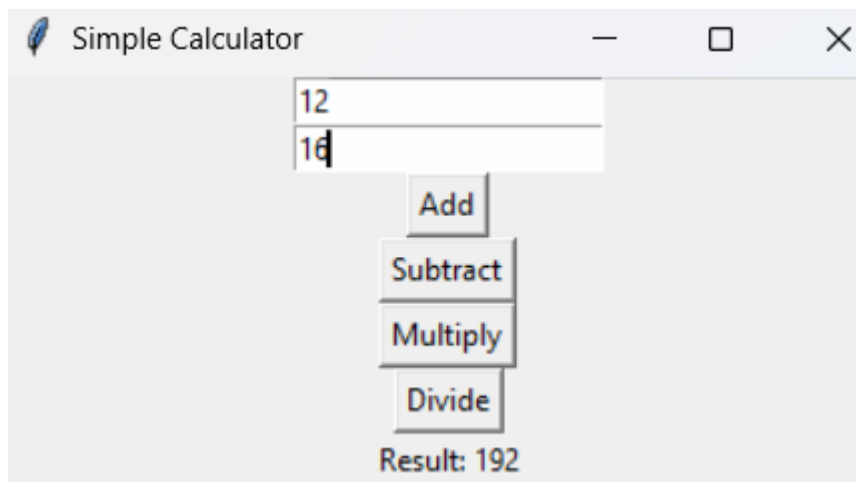
#Add



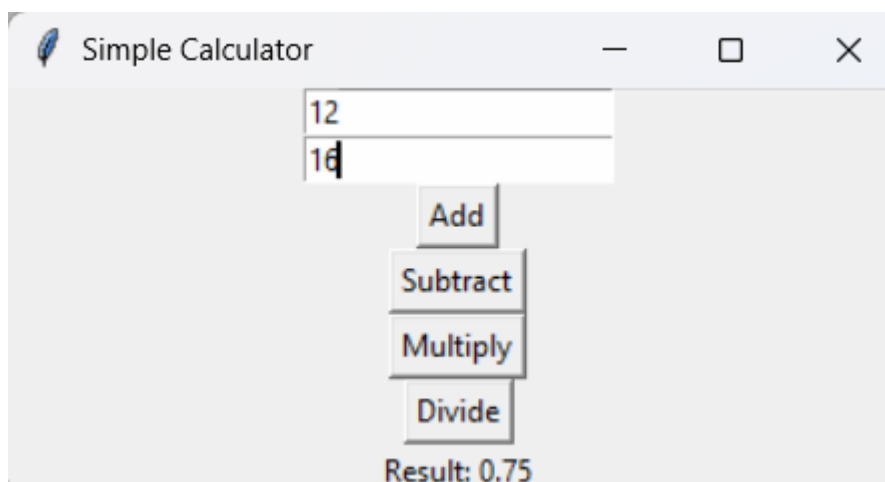
#Subtract



#Multiply



#Divide



#Task 10

```
import datetime
import random
import math
import os
import sys

print("Smart City Control Room Started")

print("Date & Time:", datetime.datetime.now())

road = random.choice(["Road A", "Road B", "Road C"])
duration = random.randint(30, 90)
print("Green signal for:", road)
print("Duration:", duration, "seconds")

usage = [120, 140, 160]
print("Average Power Usage:", math.ceil(sum(usage) / len(usage)))

file = "complaints.txt"
if not os.path.exists(file):
    open(file, "w")
print("Complaint file:", file)
print("Python Version:", sys.version)
```

```
===== RESTART: C:/Users/krsb/OneDrive/Desktop/internship/DAY 18 TASK.py =====
Smart City Control Room Started
Date & Time: 2026-02-22 20:46:45.347089
Green signal for: Road A
Duration: 83 seconds
Average Power Usage: 140
Complaint file: complaints.txt
Python Version: 3.10.0 (tags/v3.10.0:b494f59, Oct 4 2021, 19:00:18) [MSC v.1929
64 bit (AMD64)]
```

```

#Task 11

import numpy as np
import pandas as pd

print("Mini Data Analysis Project")

arr = np.array([[85,90],[78,90],[85,88]])
print("2D Array:")
print(arr)

df = pd.DataFrame(arr, columns=["Maths","Science"])
df.loc[1,"Science"] = None
df["Science"] = df["Science"].fillna(df["Science"].mean())
df = df.drop_duplicates()

print("Cleaned DataFrame:")
print(df)

print("Summary Statistics:")
print(df.describe())

```

```

===== RESTART: C:/Users/krsu/C

```

```

Mini Data Analysis Project

```

```

2D Array:

```

```

[[85 90]
 [78 90]
 [85 88]]

```

```

Cleaned DataFrame:

```

	Maths	Science
0	85	90.0
1	78	89.0
2	85	88.0

```

Summary Statistics:

```

	Maths	Science
count	3.000000	3.0
mean	82.666667	89.0
std	4.041452	1.0
min	78.000000	88.0
25%	81.500000	88.5
50%	85.000000	89.0
75%	85.000000	89.5
max	85.000000	90.0